Chelsea Kezber

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Website Portfolio • LinkedIn

SUMMARY OF SKILLS AND QUALIFICATIONS

Applications | Microsoft Office: Word • Excel • PowerPoint • MS Project • Visio • Visual Studio • LightBurn

Programming | C++ • HTML • CSS • MATLAB • Octave

Machine operation knowledge | Manual lathe and mill • CNC lathe and mill • Sheet metal bending • Grinding

CAD & Simulation| AutoCAD • CATIA • SolidWorks • ANSYS • Lotus Shark • Fusion 360 • Onshape

Languages | French | Spoken & Written • English | Spoken & Written

Licenses & Certifications | Valid driving permit • First aid license • National lifeguard license • Fire safety certificate

WORK EXPERIENCE

GCS Makerspace Laboratory Technician **July 2026 – Present**
Concordia University

- Provide technical guidance on FDM, SLA, and SLS 3D printers, Markforged printers, woodworking machinery, CNC lathe and mill, manual mill, and hand tools.
- Maintain makerspace equipment.

Mechanical Design Intern **January 2026 – September 2026**
Haply Robotics

- Created technical drawings incorporating geometric dimensioning and tolerancing (GD&T).
- Designed jigs and fixtures to support research and development activities.
- Programmed tests for R&D projects and documented results in technical reports.
- Produced prototypes using FDM, SLA, and SLS 3D printing technologies.
- Designed product packaging.
- Soldered components onto printed circuit boards (PCBs).
- Operated laser engraving equipment using LightBurn.
- Developed standard operating procedures (SOPs) for project workflows.

Engineering Design and Manufacturing Processes Lab TA **August 2025 – December 2025**
Concordia University

- Supervised students in the safe operation of manual/CNC lathes, milling machines, drill presses, and grinders, while providing instruction on machining, measurement, and assembly practices.
- Supported CAD/CAM integration (SolidWorks, Fusion 360) by assisting with toolpath generation and CNC program preparation.
- Provided technical guidance in turning, milling, drilling, tapping, and surface finishing, ensuring adherence to lab safety protocols.
- Evaluated lab reports and assignments, delivering feedback on design, manufacturing, and process documentation; collaborated with faculty on lab improvements and maintained equipment.

Conventional Machining Process Development Intern **September 2024 – December 2024**
Pratt & Whitney Canada

- Collaborated with specialists to implement new machines from development to production and optimize/modernize production lines.
- Conducted and organized experiments; analyzed castings under microscope and evaluated cutting tool wear through imaging.
- Used CATIA to model large assemblies and worked extensively with engineering drawings and strict tolerances.
- Documented manufacturing and machine processes while expanding expertise in deburring, grinding, flank/point milling, wire EDM, and fast hole EDM.
- Toured and learned from other departments, gaining hands-on exposure to casting, laser cutting, automation, metrology, and engine testing.

EDUCATION

Bachelor of Engineering – Mechanical Engineering Co-op **2023 – 2028 (Expected)**
Concordia University, Montreal, Qc

DEC in Pure and Applied Science **August 2021 – June 2023**
Dawson College, Montreal, Qc

ENGINEERING LEADERSHIP & INVOLVEMENT

Concordia Baja Racing Suspension Lead **September 2025 – Present**
SAE – Society of Automotive Engineers

- Develop suspension geometry in Lotus Shark and design components and assemblies in SolidWorks.
- Perform FEA in ANSYS to assess component stresses and deformation under design loads.
- Coordinate suspension manufacturing and integration, and develop test plans to evaluate spring rates, damping, and vehicle handling.

Concordia Baja Racing Technical Director **September 2024 – January 2026**
SAE – Society of Automotive Engineers

- Led a 40+ member team through the design, manufacturing, and development of an off-road vehicle for SAE Baja competitions.
- Reviewed engineering drawings for manufacturability and compliance with SAE Baja regulations.
- Coordinated subsystem teams, machining, and assembly to resolve integration issues and prepare the vehicle for competition.

SAE Elections Manager **March 2025 – May 2025**
SAE – Society of Automotive Engineers

- Elected by the outgoing Council to manage Concordia SAE elections, coordinating timelines, nominations, and candidate communications with the Dean of Students Office.
- Ensure fair, impartial elections that comply with university policies, announce results, and coordinate leadership transitions.

ADDITIONAL EXPERIENCE

High School Tutor **May 2024 – Present**
St. George's School

- Teach math, chemistry, physics, and technology units to students from grade 7-11.
- Promotes organizational skills, patience, communication, and time management.

Lifeguard and Swimming Instructor **April 2022 – December 2024**
Westmount YMCA

- Enhanced communication skills through frequent interactions with customers and members.
- Taught swimming to children, promoting water safety, confidence, and critical thinking skills.

INTERESTS

Karate – 1st Dan black belt, 11 years of experience, 4 years of competing in provincial and national competitions.
Sailing – CAN sail 4, competed in several sailing competitions throughout 3 years.
Cooking – I enjoy learning new techniques, recipes, and scientific reasons behind cooking; worked as a line cook for 5 years.
Designing – In my free time I enjoy designing objects and rendering them in SolidWorks.
3D Printing – 3D printing prototypes that I design on SolidWorks and Onshape